Objective:

Utilize Generative Adversarial Networks (GANs) to adeptly manipulate facial attributes of a given image, demonstrating advanced capabilities in altering age, facial expressions, hairstyle, hair colour, and facial pose.

Overview:

The main aim was to create a realistic, time-efficient, and up-to-date pipeline for facial attribute manipulation. To achieve this, a thorough review of potential conditional GAN (cGAN) models was conducted. The review considered:

- Architectural advantages
- Availability of pretrained models
- Training datasets
- Adoption and performance in literature

This guided the decision to use StyleGAN2 as the base model for conditional generation.

The candidate models were:

1. **Cycle GAN:**

    - Primarily used for **domain-to-domain transformation**.
    - Requires **separate models for each attribute/domain**, which is computationally heavy.
    - Discarded due to inefficiency for multi-attribute manipulation.

2. **StarGAN & AttGAN**

    - **StarGAN**: Single-model multi-domain translation via conditioning on domain labels.
    - **AttGAN**: Adds an attribute classification constraint at the encoder bottleneck, disentangling attributes from identity.
    - **Limitations**:

        - Pretrained models mainly on **CelebA** (low resolution, 128×128, limited attributes like pose).
        - Not widely adopted in recent research; latent manipulation approaches are now preferred.
        - Limited flexibility for new attributes without retraining.

3. **StyleGAN & StyleGAN2**

- Latent-manipulation approach rather than feature concatenation.
- Mapping network: $z \rightarrow w$ in **intermediate latent space**, where directions correspond to visual attributes (pose, hair color, age, etc.).
- **StyleGAN2** improves image quality, reduces artifacts, and produces smoother latent space geometry.
- Trained on **FFHQ dataset** (70,000 images at 1024×1024), making it standard for high-fidelity face synthesis

4. **StyleCLIP**

- Couples **StyleGAN2** with **CLIP**, a vision-language model.
- Optimizes latent codes to match a target text prompt**, with an** identity-preservation constraint (ArcFace loss).
- No retraining of the generator is needed, text guides facial edits directly.

5. **InterfaceGAN Boundaries**

- Observes that semantic attributes in **StyleGAN'**s W space are linearly separable.
- Steps:
    1. Generate large image set from StyleGAN.
    2. Label images using a pretrained attribute classifier.
    3. Train a **linear SVM** to separate attributes.
    4. Produces latent directions for attribute manipulation without training the generator.

Implementation steps:

1. Image Generation

- **StyleGAN2** model was loaded after cloning Nvidia's Pytorch's official implementation.
- Images generated using **StyleGAN2** to maintain high resolution and consistent perspective.

2. Latent Projection

- Images projected to latent space using **StyleGAN2 projector.py** (W+ space).

3. InterfaceGAN Boundaries

- Pretrained boundaries for **pose, smile, and age** were downloaded from **GenForce's InterfaceGAN** repository.
- These boundaries were used at first for facial attribute manipulation.
- Alpha values specify the amount of change for each attribute.

$$\mathcal{W}_{new} = \mathcal{W}_{old} + \alpha * boundary$$

- **Hair style** and **Hair colour** boundaries were missing so instead of retraining a model for them from scratch a new approach was explored.

4. Layer-Specific Manipulation

- Boundaries and edits applied to specific layers to avoid distorting facial structure.

5. Improved Latency with e4e Encoder

- Projecting images using StyleGAN2 projector was **time-consuming (10s of minutes)**.
- E4e encoder was downloaded from the official repository.
- Replacing the StyleGAN2's projector with **e4e encoder** reduced latency to **~3 minutes per image**.

7. StyleCLIP for Semantic Edits

- To explore further facial attributes altering the **StyleCLIP** was used, where a clone of the repo was done and the optimization technique was used to avoid training for a specific task.
- StyleCLIP was used at first to manipulate:
    - **Age**
    - **Hair colour**
    - **Hair style**
    - **Facial expressions**
    - **Pose**
- Predefined prompts tuned with **id_lambda, l2_lambda, n_steps** for optimal results.
- **Interface boundaries for Pose** produced better results than **StyleCLIP**, so they were used in the final flow.
- Dictionary used for convenience:

Attribute	Options
Hair Style	0: Mohawk, 1: Bowl cut, 2: Bob cut, 3: Curly hair
Hair Colour	4: Black, 5: Blonde, 6: Brown
Facial Expression	7: Smiling, 8: Sad, 9: Surprised, 10: Angry, 11: Frightened
Age	12: Child, 13: 20s, 14: 50s, 15: 70s, 16: 90s

Final Flow

- **StyleCLIP**: Age, hair color, hair style, facial expressions
- **InterfaceGAN boundaries**: Pose

This combination allowed **flexible, high-quality, text-driven facial attribute editing** while maintaining realism and reasonable computation time.

Experimental Results:

1. InterfaceGAN's Results:

Original Image

Altered Smile Image

Original Image

Altered Age Image

Original Image

Altered Pose Image

Original Image **Altered Pose Image**

Original Image **Altered Pose Image**

2. StyleCLIP's Result:

Original Image **A Person with Black Hair**

Original Image **A Person in their 50s**

Original Image **A Person in their 70s**

Original Image **A Child**

Original Image **A Person with Mohawk Hairstyle**

Original Image **A Person with Curly Hair**

Original Image **A Person with Bowlcut Hair**

Original Image **A Smiling Person**

Original Image **A Frightened Person**

Original Image **A Sad Person**

Original Image **An Angry Person**

Original Image **A Person with Curly Hair**

Original Image **A Frightened Person**

Original Image **A Person with Blonde Hair**

Original Image **An Angry Person**

Original Image **A Person in their 50s**

Original Image **A Surprised Person**

Original Image **A Person in their 90s**

Conclusions:

- As a baseline with no pretraining, both approaches (**StyleCLIP** and **InterfaceGAN** boundaries) showed very promising results.
- Hyperparameter tuning or continued training on the checkpoints would further strengthen model performance.
- Separate boundaries for hair color and hairstyle could be trained to study their effect on the attribute manipulation performance.
- **StyleCLIP** demonstrated superior results for age manipulation.
- The smile task performed well in both approaches.
- To extend **InterfaceGAN** boundaries for multiple facial expressions, additional training with carefully curated datasets would be required.

For the Full Implementation Steps: GitHub Repo